ICRS-ANN: AI-Driven Crop Recommendation System for Indian Agricultural Sector with SMOTE-ENN Based Balancing and TSPE Optimization

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Abstract

Purpose: Agriculture is vital to India's economy, and a crop recommendation system optimizes crop selection using soil, climate, and production data. By leveraging technology, it enhances productivity, improves resource management, reduces crop failure risk, and boosts farmers' income.

Dataset: An open use dataset from Kaggle comprising of twelve attributes are used to design the proposed crop recommendation system. The dataset is constructed from 28 years of agricultural data collected by Ministry of Statistics and Programme Implementation (MOSPI) and open government data platform india (data.gov.in).

Methodology: This article proposes an Artificial Neural Network (ANN) based crop recommendation model for Indian agricultural sector. Nine features are fed into the input layer followed by three hidden layers with dimensions 128, 64 and 32 respectively. ReLU activation function is used for introducing non-linearity and an output layer with dimension 51 (number of unique crops) is taken. After data preprocessing Synthetic Minority Oversampling-Edited Nearest Neighbor (SMOTE-ENN) is applied to address class imbalance. For efficient model training, Tree Structured Partizan Estimator(TSPE) is used for hyperparameter tuning and data is randomly sampled at 1000 samples per class. After model training ten fold cross validation technique is used to evaluate the performance of the model.

Results: The initial evaluation results 96.51% average accuracy and specificity of 99.93% without handling class imbalance and without taking random sampling. Whereas the application of SMOTE-ENN with random sampling produces enhanced average accuracy of 99.21% and specificity of 99.97%.

Keywords: Crop recommendation, SMOTE-ENN, ANN, ReLU, TSPE

1 Introduction

The agriculture sector is fundamental to India's economy, employing over 50% of the workforce, cultivating 160 million hectares, and contributing 18% to GDP [36]. However, traditional farming practices, inherited across generations, often lead to soil degradation, low yields, and inefficient resource use. The rise of hybrid crops and chemical fertilizers further exacerbates these challenges [6].

Data-driven crop recommendation optimizes productivity, ensures food security, and promotes sustainability. Advanced technologies like GIS and remote sensing enhance precision, making agriculture more resilient to climate change.

In the era of Artificial Intelligence, data-driven insights enable machine learning models to provide personalized recommendations, empowering farmers with informed decisions. This article proposes a neural network-based crop recommendation model trained on an open-source Kaggle dataset, comprising 28 years of agricultural data from MOSPI and data.gov.in. The dataset contains 99,849 entries, 12 attributes, and records for 53 major crops across four seasons in all Indian states and union territories.

The model features an input layer with 9 nodes, a hidden layer with 64 neurons using ReLU activation, and an output layer with $num_classes$ neurons representing crop classes. Data preprocessing includes outlier removal, class balancing with SMOTE-ENN, and random sampling (1,000 samples per class) to enhance generalization. Hyperparameter tuning is performed using the Tree-Structured Parzen Estimator (TSPE). The model is validated through 10-fold cross-validation, achieving 96.51% accuracy without class balancing, which improves to 99.21% with SMOTE-ENN. The primary contributions of the proposed crop recommendation system are outlined as follows:

- Key features taken are *State*, *CropType*, *Crop*, *N*, *P*, *K*, *rainfall*, *pH*, *temperature*, *Area*, and *Production*.
- Outliers were addressed in two stages: small outliers were removed using the Interquartile Range (IQR) method, while large outliers in *Area* and *Production* are handled through logarithmic transformations followed by IQR.
- The Synthetic Minority Over-sampling Technique combined with Edited Nearest Neighbors (SMOTE-ENN) is employed to manage class imbalance. KS Test is applied to validate the dataset.
- The Artificial Neural Network (ANN) is trained after utilizing Tree Structured Parzen Estimator for optimal hyperparameters, utilizing Cross Entropy Loss and the Adam Optimizer for gradient-based optimization.
- Performance is evaluated through 10-fold cross-validation, with results systematically recorded.

The structure of the remaining sections of this manuscript is outlined as follows. Some State of the art techniques for crop recommendation are discussed in Section 2. Section 3 elaborates the proposed methodology of the model. The implementation and experimental evaluation of the proposed model is performed in Section 4. Finally Section 6 draws the conclusion.

2 Literature Review

Review of status of research and development provides a knowledge foundation from the pool of the research works already proposed and it provides the motivation for the current research. Precision agriculture utilizes technology to monitor and address variability in crops within and between fields. One key area in precision agriculture is the recommendation of suitable crops. A crop recommendation system utilizing fuzzy logic was proposed by Banerjee et al. [5] which employs eight fuzzy rule-based approaches to recommend eight major crops cultivated in West Bengal, India. The system incorporates twelve parameters related to soil, terrain classification and mean annual rainfall as inputs; with the cultivation index serving as the output. The reported accuracy of this system is 92.14%. A multilayer fuzzy expert system, specifically the Mamdani fuzzy inference system, is used for recommending suitable crops as reported in [1]. The crop recommendation system incorporates twelve input variables, which are organized into six fuzzy subsystems and prioritized accordingly.

A K-nearest neighbours (KNN) based crop recommendation proposal reported in [28] have considered a large dataset containing 56 variety of crops. The model has been evaluated using different values of 'k' and achieved accuracy is more than 90% for each case. Mishra et al. [20] proposal have applied KNN and random forest over a dataset containing 22 different crops. The received accuracy is 96.96% and 98.05% respectively for KNN and Random forest respectively. Hasan et al. [12] have used agricultural "data from the relevant research institutions of Bangladesh" and have"proposed an ensemble machine learning approach, called K-nearest Neighbor Random Forest Ridge Regression (KRR)" which is a hybrid model consisting of "five existing traditional machine learning (Support Vector Regression, Naïve Bayes, and Ridge Regression) and ensemble learning (Random Forest and CatBoost)" to recommend major crops among wheat, rice and potato in aus, aman and boro season. A plenty of crop recommendation proposals have used diverse supervised learning techniques like support vector machine [35] [25], KNN [27] [14] [17]random forest [20] [29],ensemble learning [16] [4] [19] etc.

Factors like soil moister, humidity, Nitrogen (N), Phosphorous (P), Potassium (K) of the soil are influential factors for crop growth. These can be monitored by proper placement of sensors and IoT network. A research proposal in [13] "proposes an innovative ML-enabled IoT device that integrates advanced sensor technologies" to monitor soil nutrients, temperature, humidity, soil moister etc. and to recommend the type of crop to be cultivated using machine learning algorithm. A research work reported in [30] have used LDR color sensor, pH sensor, soil moister sensor, water lavel indicator, temperature sensors with Arduino UNO microcontroller to sense soil parameters and surroundings with GPS sensor to precisely indicating the area from which data are collected. The collected data are fed into "Multi-Class Support-Vector-Machine-Based Classification Model" having "linear kernel, polynomial kernel, and Gaussian kernel" with Fruit Fly Optimisation for crop and fertilizer recommendation. Rahman et al. [26] have proposed a crop recommendation model for hydroponic system

(which uses water instead of soil as the medium for growing crops) using IoT with AI. Crop recommendation is done by random forest, decision tree, KNN, Support Vector Machine, XGBoost over the dataset available from the Indian Chamber of Food and Agriculture. Some notable works on crop recommendation using machine learning with IoT (MIoT) are reported in [8] [32] [37] [33] [34].

Researchers use deep learning models for crop recommendation due to their ability to handle complex, non-linear relationships in large datasets, leading to more accurate predictions. Shingade et al. [31] have developed a recurrent neural network (RNN) in form of Long Short-Term Memory (LSTM) with the aim to crop prediction at "different places in Maharashtra". 250 sensors placed at different locations collects attributes "temperature, humidity, rainfall, and soil potential of hydrogen". The hybrid model combines the strengths of Extreme Learning Machine (ELM) and Bidirectional Long Short-Term Memory (BiLSTM). ELM performs initial feature extraction and dimensionality reduction where as BiLSTM network is trained to predict the crop to be cultivated. A hybrid model for crop recommendation in [18] has utilized multiple machine learning algorithms and LSTM. Ejigu et al. [9] have predicted rainfall and recommended the cropping pattern for Ethiopia using Radial Basis Function Neural Network (RBFNN), and K-means algorithm.

Hyperparameter tuning optimizes deep learning models for accuracy, stability, and generalization while preventing overfitting or underfitting. "To improve the overall performance" of crop recommendation model Gopi et al. [11] have taken into account Red Fox Optimization (RFO) for hyperparameter selection for deep learning models namely " long short term memory (LSTM), bidirectional LSTM (BiLSTM), and gated recurrent unit (GRU)". The model reaches an accuracy of 98.5% over an Indian dataset. Venkatanaresh and Kullayammas' proposal [38] has considered "Concurrent Excited Gated Recurrent Unit (CE-GRU) based DL approach" and " hunter-prey optimization (HPO) technique" for hyperparameter tuning. The model achieves an accuracy of 99.2% under learning rate 80% over an dataset from Maharashtra having 12,629 entries. Few other notable works related to crop recommendation using deep learning have used ant colony optimization [21], sea horse optimization [3] etc. for hyperparameter tuning.

3 Methodology

The proposed research focuses on crop recommendation within the Indian agricultural sector using a dataset spanning 28 years from the Ministry of Statistics and Programme Implementation (MOSPI) and the Open Government Data Platform (data.gov.in). The dataset undergoes preprocessing, including missing value imputation, outlier handling, and class balancing using SMOTE or Random Sampling. Feature selection is performed via correlation analysis, followed by label encoding for categorical features and standard scaling for optimal model training. The preprocessed data is then fed into a neural network with nine input

nodes, 64 hidden nodes, and c output nodes, where c represents the number of crops. The model's workflow is depicted in Fig 1.

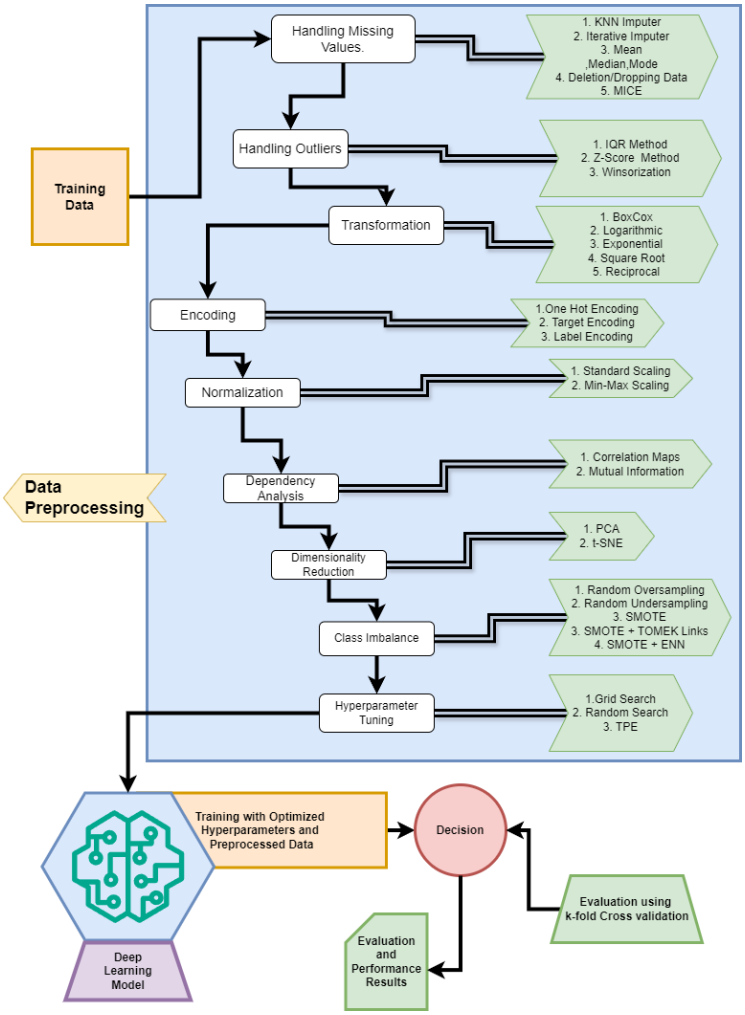


Fig. 1: Flow Diagram of Crop Recommendation System Methodology.

3.1 Dataset

The crop recommendation model is built upon a comprehensive dataset sourced from Kaggle [24]. As mentioned by the author the dataset is a collection of twenty eight years of agricultural data from Ministry of Statistics and Program Implementation (MOSPI) and open government data platform India (data.gov.in). The dataset contains 99,849 entries covering 53 crops from the 33 States and

union territories in 'rabi', 'kharif', 'summer' and 'whole year' seasons. The features includes 'State name', 'crop type (season)', 'crop', 'N (nitrogen in kg/hec)', 'P (phosphorous in kg/ hec)', 'K (potassium in kg/hec)', 'pH', 'rainfall', 'temparature' , 'area in hector', 'production in tons', 'yield ton per hec'. It is to be noted that no missing values is noticed in any of the attributes. Additional details are present in Supplementary Material Fig and Table 1

3.2 Data preprocessing

Data preprocessing is a crucial step in preparing the dataset for various machine learning and deep learning models. The step can be used to handle missing values, handling outliers, transformations, encoding, scaling, and addressing class imbalance. Each technique is utilized to enhance the quality of the dataset and suitability for model training.

(I) **Handling of missing values:** Datasets often contain missing values due to factors like human error, sensor failure, or systematic data collection gaps. Various techniques like KNN Imputer, Iterative Imputer, Multivariate Imputation with Chain-Equation (MICE) have been proposed to address this issue [10] [23].

(II) **Handling outliers:** Outliers are data points that significantly deviate from the overall data pattern, potentially skewing analysis and leading to misleading conclusions. In machine learning, outliers can distort predictions, increase errors, and reduce generalization, causing models to overfit noisy data and compromising accuracy, precision, and robustness. Detecting and handling outliers is essential for model reliability and performance and can be handled by using techniques like IQR, Z-Score, Winsorization etc. In this reserach IQR method as discussed below is used for preprocessing.
IQR Method: The Interquartile Range (IQR) method is a widely used technique for detecting and removing outliers. It calculates IQR as the difference between the 75th percentile (Q3) and the 25th percentile (Q1). Data points falling outside the range $Q1 - 1.5 \times IQR$ and $Q3 + 1.5 \times IQR$ are considered outliers and handled accordingly.Here Q_1 and Q_3 signifies median of lower half and upper half of the data respectively and IQR= $Q_3 - Q_1$.

(III) **Transformation**: Transformations stabilize variance and make data more normally distributed. Various techniques, such as log transformation, Box-Cox transformation, square transformation, square root transformation, and exponential transformation, are applied based on the dataset's distribution. For example, log transformation reduces skewness, Box-Cox works for both positive and negative skewness, and square transformation enhances larger values, increasing data spread.

(IV) **Encoding**: Encoding categorical values is a crucial preprocessing step that converts categorical data into a numerical format for algorithmic processing. Various encoding techniques exist, each suited to specific categorical variable types. The choice of technique depends on the data and model requirements. Common encoding methods like One-Hot encoding, Target encoding, Label

encoding etc. are commonly used. A brief discussion of Label encoding is presented below.

Label Encoding: Label encoding is a simple technique that converts categorical variables into numerical form, enabling machine learning algorithms to process them. Each unique category in a feature is assigned a distinct integer, typically based on alphabetical sorting or natural ordering if applicable. Given a categorical variable X with k distinct categories $\{x_1, x_2, \ldots, x_k\}$, each category x_i is assigned a unique integer label l_i where $l_i = i$ for $i = 1, 2, \ldots, k$. Each data point $x \in X$ is then replaced with its corresponding integer label l_i.

(V) **Normalization**: Normalization in deep learning ensures input features are on a compatible scale, enhancing training efficiency and effectiveness. Common techniques like standard scaling and min-max scaling promote stable and faster convergence, reduce numerical instability, and improve model performance and generalization. Techniques like Min-Max Scaling, Standard Scaling etc are generally utilized. A Brief discussion of Standard Scaling is presented below.

Standard Scaling:

Standard scaling (or Z-score normalization) standardizes features by removing the mean and scaling to unit variance, ensuring the data has a mean of 0 and a standard deviation of 1. Given a dataset $\{x_1, x_2, \ldots, x_n\}$, the mean is computed as $\mu = \frac{1}{n}\sum_{i=1}^{n} x_i$, and the standard deviation is given by $\sigma = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_i - \mu)^2}$. Each data point x_i is then transformed using $z_i = \frac{x_i - \mu}{\sigma}$, resulting in a standardized dataset $\{z_1, z_2, \ldots, z_n\}$ with a mean of 0 and a standard deviation of 1. However, standard scaling can be sensitive to outliers, which may distort the mean and standard deviation, leading to skewed results.

(VI) **Dependence Analysis**: Dependency analysis helps uncover relationships between variables, aiding in feature selection, dimensionality reduction, and predictive modeling. Two key techniques are correlation maps and mutual information. Correlation maps visually represent pairwise correlation coefficients, highlighting linear dependencies, while mutual information quantifies shared information between variables, capturing both linear and non-linear relationships. Correlation maps are simple and computationally efficient, whereas mutual information is more robust for detecting complex dependencies. Techniques like Correlation Maps, Mutual Information etc. are utilized. A Brief discussion of Correlation Maps is presented below.

Correlation Maps: Correlation maps visually represent the pairwise correlation coefficients between variables, helping to identify linear dependencies.

Pearson Correlation Coefficient: It is a statistical measure (see Eqn 1) that quantifies the linear relationship between two continuous variables. The value ranges from -1 to +1, where +1 indicates a perfect positive correlation, -1 signifies a perfect negative correlation, and 0 represents no linear

correlation between the variables.

$$\rho_{X,Y} = \frac{\text{cov}(X,Y)}{\sigma_X \sigma_Y} \tag{1}$$

where

$$\text{cov}(X,Y) = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y})$$

In this context, $\bar{X}$ and $\bar{Y}$ represent the average values of X and Y respectively, while n denotes the total number of observations. The standard deviations for X and Y are denoted by σ_X and σ_Y respectively..

(VII) **Dimensionality Reduction**: Dimensionality reduction streamlines datasets, improving analysis and model performance by reducing input variables while retaining essential information. This enhances storage efficiency and interpretability during training. techniques like PCA, t-SNE etc are utilized . A brief discussion of PCA is presented below.

Principal Component Analysis (PCA) Principal Component Analysis (PCA) transforms high-dimensionaldata into a new coordinate system where the maximum variance aligns with the initial principal components. It converts the data into orthogonal components ranked by explained variance, capturing the most significant patterns. The process begins by standardizing the dataset, where each feature is centered by subtracting the mean $\bar{\mathbf{x}}$ and divided by its standard deviation σ, yielding $\mathbf{z} = \frac{\mathbf{x}-\bar{\mathbf{x}}}{\sigma}$. Next, the covariance matrix $\mathbf{C}$ is computed as $\mathbf{C} = \frac{1}{n-1}\sum_{i=1}^{n}(\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^T$. The eigenvalues λ and eigenvectors $\mathbf{v}$ of $\mathbf{C}$ are then determined by solving $\mathbf{Cv} = \lambda\mathbf{v}$. The eigenvalues and corresponding eigenvectors are sorted, and the top k eigenvectors are selected to form matrix $\mathbf{W}$. Finally, the dataset is transformed using $\mathbf{Y} = \mathbf{XW}$, producing the reduced-dimensional representation of the original data.

(VIII) **Class Imbalance**: Class imbalance occurs when one or more classes are significantly underrepresented compared to others. Addressing this issue is crucial to prevent model bias toward the majority class, ensuring better performance for minority classes. Various techniques can be used to balance the dataset and enhance model generalization. Techniques like Random Oversampling, Random Undersampling, SMOTE, SMOTE+Tomek Links , SMOTE-ENN etc. are utilized. A Brief discussion of SMOTE-ENN is presented below.

SMOTE + ENN (Edited Nearest Neighbors): This technique first applies SMOTE to generate synthetic minority instances and then uses Edited Nearest Neighbors (ENN) to remove instances based on nearest neighbor rules. Given a dataset D with a minority class M and a majority class N, SMOTE is applied to M to generate a new set of synthetic instances M'. The dataset is then updated as $D' = M' \cup N$. A parameter k is set to define the number of nearest neighbors. For each instance x in D', the set of k nearest neighbors N_k is identified, and the majority class label y within N_k is determined. If x belongs to the minority class and its label differs from the

majority class label, it is removed from D'. The final balanced and cleaned dataset D' is returned, enhancing class boundary definition and reducing noise.

(IX) **Hyperparameter Tuning:** Hyperparameter tuning is essential for optimizing machine learning models by selecting the best hyperparameters to enhance performance. Common methods include Grid Search, Random Search, and Tree-Structured Parzen Estimator (TSPE). A Brief discussion of TSPE is presented below.

Tree Structured Parzen Estimator: The Tree-Structured Parzen Estimator (TSPE) [7] (see Algorithm 1) is an advanced hyperparameter optimization method using Bayesian optimization. Unlike Grid and Random Search, TSPE constructs probabilistic models to pinpoint promising areas within the search space, efficiently selecting configurations likely to improve performance. By leveraging models like Gaussian mixtures, it reduces evaluations while optimizing deep learning and other computationally expensive algorithms.

Algorithm 1 Tree-Structured Parzen Estimator (TSPE)

Input: Hyperparameter space H with parameters $h_1, h_2, \ldots, h_k$, number of iterations n

Output: Best hyperparameters h^*

Initialize history of trials $T \leftarrow \emptyset$(NULL).
for $i = 1$ to n **do**
    Fit two models: $l(h)$ and $g(h)$ using T to estimate the likelihood of good and bad hyperparameters.
    Calculate the acquisition function $\alpha(h) = \frac{l(h)}{g(h)}$.
    Select hyperparameters h that maximize $\alpha(h)$.
    Train the model using hyperparameters h.
    Evaluate the model to obtain score s.
    Update the history $T \leftarrow T \cup \{(h, s)\}$.
end for
Return the hyperparameters h^* with the best score in T.

3.3 Model Building and Training Procedure

3.3.1 Model architecture Artificial Neural Networks (ANNs) are nonlinear learning models inspired by the human brain, consisting of interconnected layers of neurons. They recognize patterns, classify data, and aid in prediction and decision-making. The proposed recommendation model utilizes a Multilayer Perceptron (MLP), a common ANN for handling nonlinear datasets, comprising input, hidden, and output layers. Designing an optimal network structure remains challenging, as no systematic method exists [2]. Researchers often rely on trial and error, adjusting hidden layers and units to minimize error and maximize accuracy. Each neuron in the input layer receives data directly from the

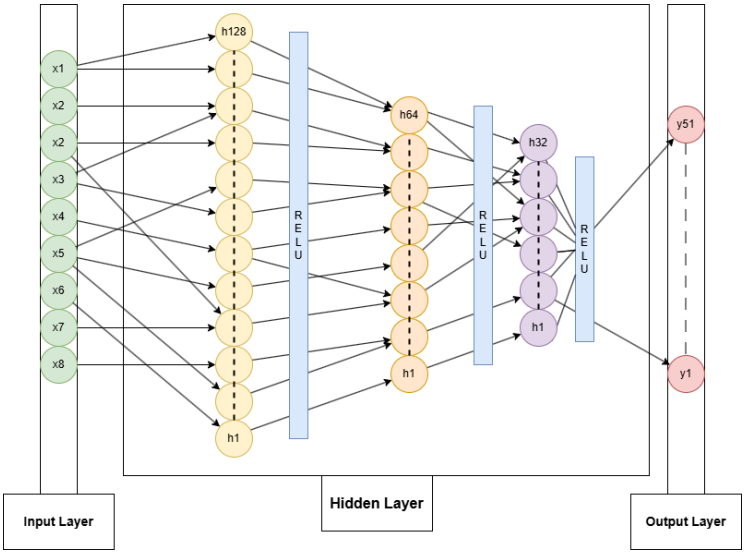


Fig. 2: Model Architecture.

dataset, and each input signal is then transmitted to all neurons in the hidden layer. The neuron in the hidden layers computes the weighted sum received from preceding layer and activation function (e.g., sigmoid, ReLU, tanh) is applied over it as presented in Eqn 2 .

$$y = f(z) = f(\sum_{i=1}^{n} w_i x_i + b) \tag{2}$$

where x_i are the inputs, w_i are the associated weights, b is bias and f is the activation function.

Activation functions introduce non-linearity, allowing the network to learn complex patterns. Common functions include Sigmoid, ReLU, and Softmax, with ReLU used in the proposed model.

The ANN consists of one input layer, three hidden layers, and one output layer. The input layer has nine neurons (explained in Section 4). The hidden layers contain 128, 64, and 32 neurons, respectively, while the output layer has 51 classes. Among 53 unique crops in the dataset, 'apple' and 'coffee' are excluded due to insufficient records (2 and 1, respectively). The model architecture is shown in Fig 2. The ReLU function is defined in Eqn 3:

$$\text{ReLU}(x) = \max(0, x) \tag{3}$$

ReLU is chosen as it mitigates the vanishing gradient problem [22], enabling faster convergence and improving training efficiency. The hidden layer neurons are set in multiples of 32 to optimize GPU performance, ensure memory alignment, and enhance compatibility with deep learning libraries.

3.3.2 Training Procedure For the training purpose the dataset is partitioned into a ratio of 80:20 for training and testing. The aforementioned model is trained using the preprocessed data by Adam(Adaptive Moment Estimation) optimizer [15] which modifies the learning rate throughout the training procedure.

$$L = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} y_{ic}\log(\hat{y}_{ic}) \tag{4}$$

where L is defined as the average negative log-likelihood of the true class labels, N represents the number of samples in the dataset and C denotes the number of classes. The ground truth label y_{ic} is a one-hot encoded value, where $y_{ic} = 1$ if the sample i belongs to class c, otherwise $y_{ic} = 0$. The predicted probability $\hat{y}_{ic}$ is the model's estimated likelihood that sample i belongs to class c, typically obtained through a softmax function. The term $\log(\hat{y}_{ic})$ represents the natural logarithm of this predicted probability. Cross Entropy Loss (Eqn 4) is used during backpropagation to compute gradients and update weights across training epochs.In the training phase, hyperparameters include the number of epochs and the initial learning rate. The Tree-Structured Parzen Estimator (Algorithm 1) is employed to optimize these values efficiently, preventing resource wastage.To address class imbalance and prevent overfitting, 1000 randomly sampled records from each output class are used during training.

4 Implementation and Evaluation

As mentioned in section 1 an opensourced dataset from Kaggle is used for the implementation purpose. The dataset is passed through preprocessing, training and evaluation. Following subsections present the implementation steps in details.

4.1 Preprocessing

As depicted by Fig. 1, nine different techniques are applied to preprocess the dataset.

(i) Missing Values:- The Dataset has no instances of having missing values (See Supplementary Material Fig. 2) ,hence preprocessing steps regarding handling missing values are not necessary.

(ii) Outliers and Transformation:- Outliers are detected using box plots (see Figure 3). Slight skewness in N, P, K, pH, Rainfall, and Temperature is handled using the IQR method. For highly skewed *area_in_hectares*, *production_in_tons*, and *yield_ton_per_hec*, log transformation followed by IQR is applied (see Figure 4).

(iii) Class Distribution:- A class distribution graph (Supplementary Material Figure 3) shows the frequency of crop labels in the dataset. 'Apple' and 'Coffee' have the lowest occurrences (2 and 1, respectively) and are discarded due to insufficient data, resulting in 51 output classes for the recommendation system.

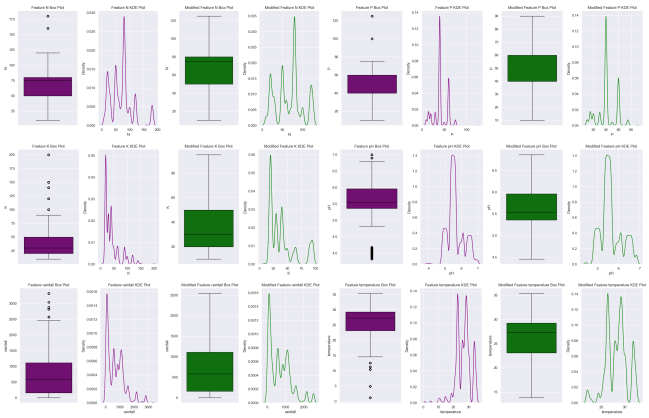


Fig. 3: Box Plots and KDE Plots for Before and After IQR Method.

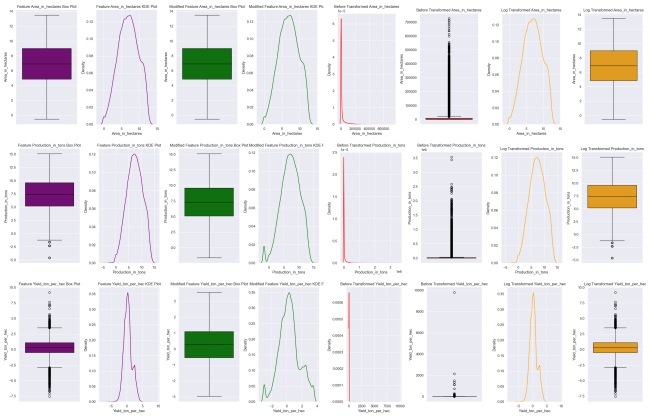


Fig. 4: Box Plots and KDE Plots for Log Transformation and IQR Method, Before and After.

(iv) Enconding: The categorical features in the dataset are encoded using Label Encoding, transforming each unique category into a numeric label to allow our model to process and learn from these features efficiently. An snap sort of the dataset after encoding is shown in Supplementary Material Fig. 4

(v) Normalization:- Normalization ensures the input features to be on a compatible scale, which enhances the efficiency and effectiveness of the training process. Except the 'Crop' target feature every other features undergo Standard scaling. Supplementary Table 2 presents the ranges of the features before and after scaling.

(vi) Dependency Analysis:- To find the dependency among the features, correlation plot (see Supplementary Material Fig. 5) are constructed. This is

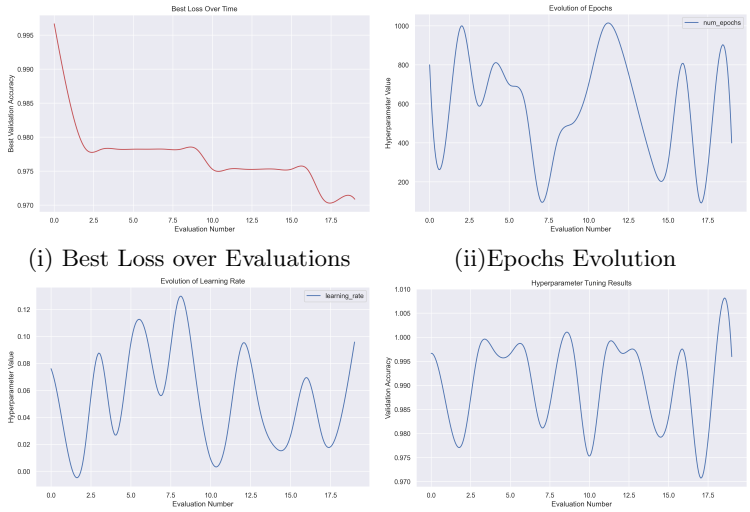
done by merging highly correlated features or removing one of them from the dataset.

(vii) Dimensionality Reduction:- The correlation plot in dependency analysis identifies *Area_in_hectares* and *production_in_tons* as highly correlated (0.88). Since both are crucial for predicting the target feature, they are combined using PCA to prevent data loss.

(viii) Class Imbalance:- Class imbalance affects training performance, so balancing techniques are applied. SMOTE-ENN is chosen for optimal results after evaluating multiple techniques (see Supplementary Material Fig. 13, 14, 15, 16 for training results and Table 3, 4, 5, 6, 7 for evaluation outcomes. see Supplementary Material Fig. 6 for KS Test).

4.2 Training

4.2.1 Hyperparameter Tuning Hyperparameter tuning was conducted using TSPE (Algorithm 1) with epochs and learning rate as the search space. Figure 5(i) illustrates loss evolution, 5(ii)epoch progression, 5(iii) learning rate changes, and 5(iv)maximum training accuracy over 20 evaluations. Results from other techniques are in Supplementary Material Fig. 7, 8, 9, 10 and 11.



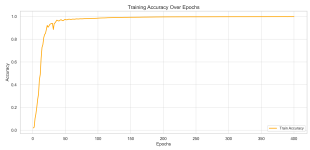
(i) Best Loss over Evaluations (ii)Epochs Evolution

(iii)Learning Rate Evolution (iv)Training Accuracy Evolution

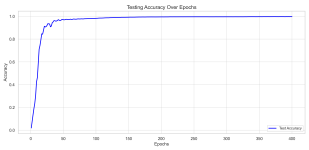
Fig. 5: Hyperparameter Tuning Results

4.2.2 Training and Result The optimal hyperparameters from Section 4.2.1 are used to train the proposed model (Section 3.3.1). Figures 6(i, ii) illustrate

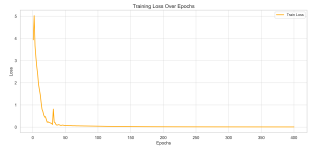
training/testing accuracy, 6(iii, iv) depict loss trends, and overall model performance is presented in Fig. 6 (v, vi). The ROC-AUC curve evaluates class distinction, with AUC closer to 1 indicating better performance. The radar chart (Figure 6 (vi)) visualizes Precision (minimizing false positives), Recall (capturing actual positives), and the F1 score (balancing both). See Supplementary Material Fig. 12, 13, 14, 15 and 16 for results using other techniques.



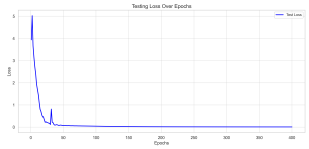
(i) Training Accuracy over Epochs



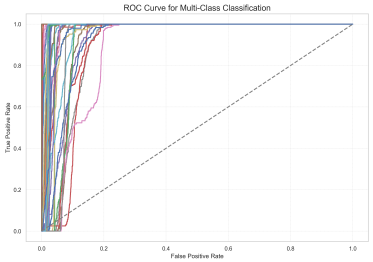
(ii) Testing Accuracy over Epochs



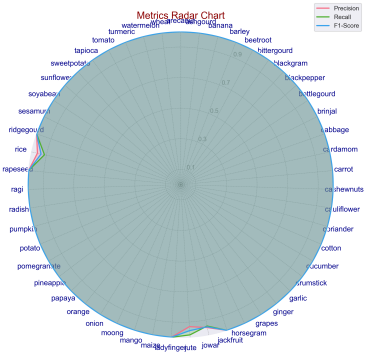
(iii) Training Loss over Epochs



(iv) Testing Loss over Epochs



(v) ROC AUC Score



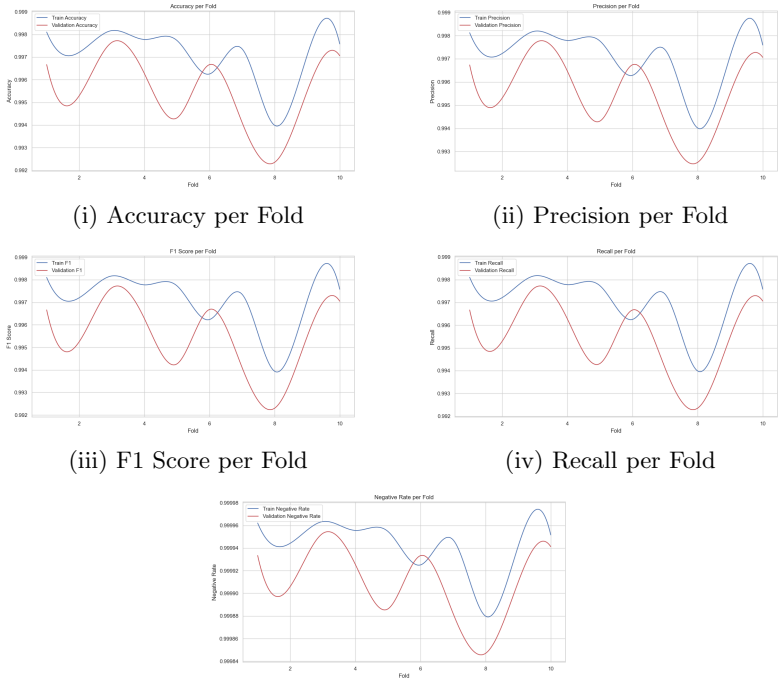
(vi) Radar Chart

Fig. 6: Combined Graphs: Loss, Accuracy, ROC and Radar Chart

4.3 Evaluation

The model is assessed using 10-fold cross-validation, yielding key metrics: accuracy, precision, recall, F1-score, and negative rate. Figure 7 summarizes the results, where Fig. 7(i) presents accuracy across folds, Fig. 7(ii) depicts precision variations, Fig. 7(iii), (iv), (v) illustrate F1-score trends, recall, and negative rate respectively. The results indicate stable performance with minimal variance,

confirming the model's reliability for crop recommendation. See Supplementary Material Table 3, 4, 5, 6, 7 for results using other techniques.



(i) Accuracy per Fold

(ii) Precision per Fold

(iii) F1 Score per Fold

(iv) Recall per Fold

(v) Negative Rate per Fold

Fig. 7: Evaluation Results Across Folds: Accuracy, Precision, F1 Score, Recall, and Negative Rate

5 Comparison

The proposed technique is compared with some recent state-of-the-art methods over the same field on different approaches(See Table 2). It is clear from the table that the proposed method gives the best result in terms of accuracy.

6 Conclusion

The proposed work presents a crop recommendation system using Artificial Neural Network (ANN) over an Indian dataset. SMOTE-ENN is employed to handle the class imbalance in the dataset and the enhanced dataset is evaluated

Fold	Train Loss	Train Acc.	Test Loss	Test Acc.	Train Prec.	Train Rec.	Train TNR	Train F1	Test Prec.	Test Rec.	Test TNR	Test F1
1	0.008926	0.997	0.014526	0.996	0.997	0.997	1.000	0.997	0.996	0.996	1.000	0.996
2	0.007250	0.998	0.007971	0.997	0.998	0.998	1.000	0.998	0.997	0.997	1.000	0.997
3	0.005922	0.998	0.009591	0.997	0.998	0.998	1.000	0.998	0.997	0.997	1.000	0.997
4	0.008284	0.998	0.011202	0.996	0.998	0.998	1.000	0.998	0.996	0.996	1.000	0.996
5	0.014247	0.995	0.018340	0.993	0.996	0.995	1.000	0.995	0.993	0.993	1.000	0.993
6	0.005961	0.998	0.010676	0.996	0.998	0.998	1.000	0.998	0.997	0.996	1.000	0.996
7	0.007158	0.997	0.014377	0.995	0.997	0.997	1.000	0.997	0.995	0.995	1.000	0.995
8	0.007280	0.998	0.014976	0.997	0.998	0.998	1.000	0.998	0.997	0.997	1.000	0.997
9	0.005663	0.998	0.012752	0.997	0.998	0.998	1.000	0.998	0.997	0.997	1.000	0.997
10	0.005306	0.998	0.007382	0.998	0.998	0.998	1.000	0.998	0.998	0.998	1.000	0.998
Average	0.007600	0.998	0.012179	0.996	0.998	0.998	1.000	0.998	0.996	0.996	1.000	0.996

Table 1: Performance metrics across 10 folds with test metrics

Source	Proposed Method	Performance Metric
Bandi et al. [4]	Ensemble model with majority voting	99.4% Accuracy
Islam et al. [13]	CatBoost Classifier	97.5% Accuracy
Senapaty et al. [30]	Multi-class support vector machine using directed acyclic graph and fruit fly optimisation	97.3% Accuracy
Rahman et al. [26]	Random Forest algorithm	97.5% Accuracy
Sravanthi et al. [34]	Multi-level Kronecker Guided Pelican Convolutional Neural Network (MKGPCNN)	99% Accuracy
Gopi et al. [11]	Red fox optimization with ensemble recurrent neural network	98.45% Accuracy
Ashoka et al. [3]	Enhanced top-down attention spiking neural network	98.94% Accuracy
Proposed Work	ANN with SMOTEENN and TSPE	99.6% Accuracy

Table 2: Comparison of related works

using Kolmogorov-Smirnov (KS) test. Hyperparameter tuning is performed using the Tree-Structured Parzen Estimator, optimizing the model's performance for robust and reliable predictions. Rigorous k-fold cross-validation confirms the system's robustness and generalizability, ensuring consistent and accurate recommendations. Performance metrics, AUC-ROC Curve, Radar Chart signify the robustness of the proposed model. The crop recommendation system provides valuable data-driven insights for optimizing crop selection and boosting agricultural productivity. Future work may involve integrating additional features and utilizing more sophisticated deep learning algorithms to further refine the system's performance.

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